3 Yr. Degree/4 Yr. Honours 3rd Semester Examination, 2024 (CCFUP)

Subject : Computer Science

Course: COMP3011 (MAJOR)

(Discrete Structures)

Time: 3 Hours **Full Marks: 60**

The figures in the right hand margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

1. Answer *any ten* of the following questions: 2×10=20

(a) If a set B has n elements, then what is the total number of subsets of B? Justify your answer.

(b) If A = { 3, 5, 7, 9, 11}, B = {7, 9, 11, 13}, C = {11, 13, 15}, find $A \cap (B \cup C)$, $A - B$, $B - C$.

(c) Differentiate Regular graph and Complete graph.

(d) Simplify the expression $(x + y)(x + z)$ using the laws of Boolean algebra.

(e) Find out the number of ways that the letters of the word "LEADER" can be arranged?

(f) Define closed walk in a graph with example.

(g) Define Tautology and Contradiction. Give examples.

(h) Write the applications of Bipartite graph.

(i) Draw the graphs W_7 and Q_4.

(j) Define simple and compound propositions with examples.

(k) If G is a simple graph with 15 edges and $\overline{G}$ has 13 edges, how many vertices does G have?

(l) "*The chromatic number of* $K_n = n$".— Justify the statement.

(m) Define Modus Ponens rule.

(n) A simple graph G has 24 edges and degree of each vertex is 4. Find total number of vertices.

(o) What is Recurrence relation?

2. Answer *any four* of the following questions: 5×4=20

(a) Prove that a simple graph with n vertices must be connected if it has more than [(n-1)(n-2)]/2 edges.

(b) Construct truth table for (1) $p \vee q \oplus p \wedge q$, (2) $(p \vee q) \vee r$

(c) Show these are logically equivalent

(i) $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$

(ii) $p \leftrightarrow q$ and $\neg p \leftrightarrow \neg q$

(d) Show that $1^2 + 2^2 + \ldots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$, $n \geq 1$ by Mathematical Induction.

(e) Find the time using substitution method for the following: $T(n) = T(n-1) + n$, $n > 1$

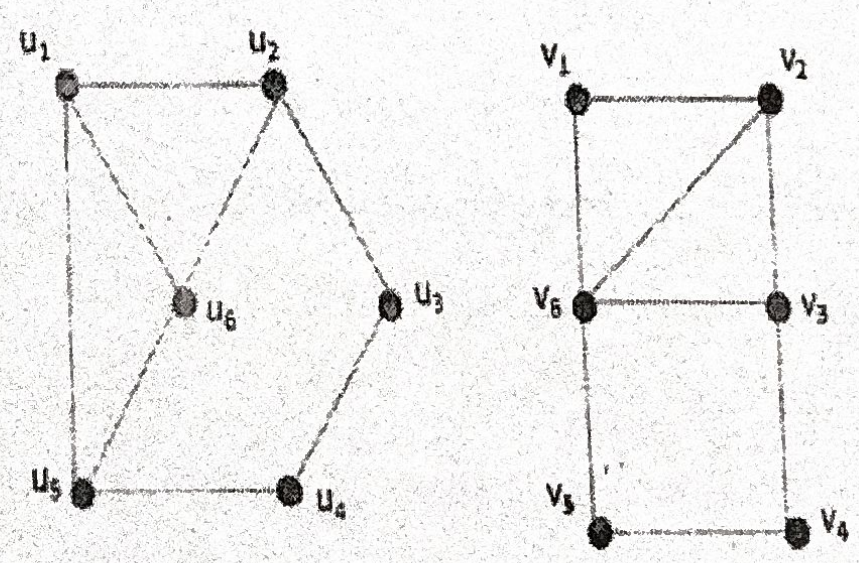
(f) In a room there are 10 people, none of whom are older than 50 but each of whom is at least 5 years old.

Prove that we can always find two groups of people (with no common person) the sum of whose ages is the same.

3. Answer *any two* of the following questions: 10×2=20

(a) State and prove De Morgan's laws. Prove that $\neg q \wedge (p \rightarrow q) \rightarrow \neg p$ is a tautology.

(b) What is Euler Circuit? "Every tree with two or more vertices is 2-chromatic". — Justify the statement. Determine whether the given pair of graphs is isomorphic.



(c) (i) Solve the summation: $\sum_{n=2}^{60}(2n^2+3)$

(ii) The limit $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{3}{n}\left(\frac{n+k}{n}\right)^2$ is obtained by applying the definition of the integral to $\int_1^2 f(x)dx$. What is the function of $f(x)$? 4+6

(d) (i) Solve: $a_n = a_{n-1} + a_{n-3}$, $a_0 = 1$, $a_1 = 2$, $a_2 = 0$ for $n = 6$. 5+5

(ii) Derive the recurrence relation for Fibonacci Series.

$a_6 = a_5 + a_3$

$a_5 = a_4 + a_2$

$a_4 = a_3 + a_1$

$a_3 = 2$

$a_{n-1} = a_{n-2} + a_{n-4}$

$a_{n-3} = a_{n-4} + a_{n-6}$

$x^2(x-1)$

$x^2(x-1)$

$a_5 - k$

$x^3 - x^2 - x + 3$

2

$-1 \quad -1 \quad -1$

$a \quad a_0 = 4 - 1$